Supplementary file 4. Characteristics of included studies

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Aihara, 2006 Thailand	Cross-sectional study	One district in Kanchanburi province, Thailand	Mothers	224	MCH handbook	No comparison group	There was a low rate of reading (14.3% mother had read all of the contents) and self-recording (0.9% mother had recorded every part). Multiple regression coefficients showed utilization of the MCH handbook was related to both mother's MCH promoting belief (*p*-value=0.001) and action (*p*-value=0.039). This was the strongest predictor variable of mother's MCH promoting belief. Other factors which significantly related to MCH promoting belief were family income, age, and education, and relation to action were marital status, occupation and age.
Akiba, 2016 Japan	Cross-sectional	College of Education, Ibaraki University, Ibaraki, Japan	Female university students between 18-22 years of age whose parents also provided consent to participate in the study	41	MCH handbook	Those who did not receive or record in MCH Handbook	Personal records written in the MCH Handbook could be a predictor of school maladaptation. Children of mothers who wrote at least one record of worrying/anxious behavior in the MCH Handbook were more likely to develop maladaptation in school environment (*p*-value<0.05).
Aoki, 2009 Japan	Cross-sectional	Three nursery schools in Tokyo and one nursery school in Saitama Prefecture	Parents of nursery school students (0-5 years old)	298	MCH handbook	No comparison group	Checking of developmental milestones at various time points was frequent, but recording of growth curves or observations of children was done less often. Lnformation in the MCH handbook was not used frequently. In general, guardians used the handbook passively rather than actively, and only about half regarded the handbook as user-friendly. To improve the quality of the MCH handbook, guardians requested more information on child health, such as first aid, the timing of immunization, or weaning foods. On the basis of categorical data analysis of the results, a "user-friendly MCH handbook" was considered to incorporate the following points: an appropriate size, easy-to-understand expressions, and a higher content of information relevant to guardians.
Bhuiyan, 2006 Bangladesh	Mixed methods	Maternal and Child Health Training Institute in Dhaka, Bangladesh	Pregnant women	600	MCH handbook	Standard cards	Findings from the focus group discussions emphasized the need for including MCH handbook in maternal and child program in Bangladesh. In addition, quantitative data suggests that mothers in study group had higher knowledge on MCH issues, better practices in MCH care, and higher utilization of MCH services than mothers in control groups who used other health cards.
Clendon, 2010 New Zealand	Qualitative	New Zealand Plunket society	Mothers who used the plunket book	35	Plunket book	No comparison group	The book plays an important role in the relationship between mother and nurse. It is used as a point of commonality that supports the efforts of both as they work toward establishing an effective relationship, as a tool of practice, and as a means of building strength within families.
Du Plessis, 2017 South Africa	Cross-sectional	143 PHC facilities across all six health districts in Western Cape Province	Children between the ages of 0 and 36 months	5,193	Road-to-Health-Booklet	No comparison group	All healthcare workers indicated that health promotion messages were important. However, messages were only conveyed in 51% of observed consultations. When it was communicated, health promotion messages were age-appropriate in 97% of cases. Barriers to the implementation of health promotion messages hinged on time and staf constraints, workload and language barriers. Various forms of health promotion material were available in facilities.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Elbourne, 1987 UK	RCT	Peripheral consultant clinic in Newbury, West Berkshire	Expectant mothers	290	Maternity case notes (full records)	Standard cooperation card (abbreviated version of the full obstetric record)	Women holding their full records were significantly more likely to feel in control of their antenatal care (RR [Rate Ratio] =1.45, 95% CI: 1.08-1.95) and to feel it was easier to talk to doctors and midwives. No other beneficial effects were detected. Women holding their own records were more likely to say that they would prefer the same kind of record again in a subsequent pregnancy than were women holding a cooperation card (RR=1.56, 95% CI= 1.34-1.81). Women holding their case notes did not feel more anxious than cooperation card holders.
Engida, 2013 Ethiopia	Qualitative	Amhara region, Ethiopia	Health extension workers, health development army members, care takers (breast feeding mothers and pregnant women)	112	Speaking books	No comparison group	Speaking Books were perceived well by the health extension workers and health development army members, and it was agreed that it is an effective tool to disseminate information.
Fujii, 2020 Japan	Qualitative	Social club for mothers of twins held in Tokyo	Mothers of twins	5	MCH handbook	No comparison group	Mothers who had given birth to twins regarded the MCH handbook as "evidence of their readiness to become mothers of twins," "hope of becoming a good mother," something that should prevent anxiety related to having a high-risk pregnancy," "a medical record that shows how the child is developing" and "they stopped using the handbook on their own."
Fujimoto, 2001 Japan	Cross-sectional	231 local towns and wards in Niigata, Yokohama, Shizuoka, and Hiroshima	Caregivers who have come with their 18-month old child for 18-month check-up	10,900	MCH handbook	No comparison group	High ratio of caregivers who read and wrote in the MCH handbook. Loss was minimal at 0.9%. The most responses for the most useful page was the "vaccination record". Many expected to see improvements in "child rearing" information. Many caregivers replied in neutral when asked about the usefulness of the handbook. Oral hygiene was the least filled-out and there was only a minimum of people who replied that this page was useful.
Gholipour, 2018 Iran	Cluster RCT	21 health centers and health posts in Tabriz	Pregnant women	185	Maternity books with group support sessions	Standard care and no maternity books	The intervention showed positive effects on the service quality and customer quality of maternity care through increased involvement of mothers in the care process.
Grippo, 2007 Brazil	Mixed methods	Family Health Program in the city of Sao Paolo	Family caregivers responsible for 0–59-month-old children	89	Educational booklet *Toda hora e hora de cuidar* (Anytime is time to care)	No comparison group	Even though mothers had not completed basic education, they reported the booklet contents were understandable and interesting. The concept regarding childcare was related to affective and work activities. The booklet is effective as an instrument to promote skills and potentials of the community, family, and individuals.
Grøvdal, 2006 Norway	RCT	Maternal and child health centers in 10 municipalities in Norway	Parents of 309 children attending the National Preschool Health Surveillance Programme	309	Half of the parents were given a parent-held child health record (PHCHR) and short instructions on how it was expected to be used.	Parents and children who did not use PHCHR, just ordinary national health surveillance program	Some 73% of the intervention group used the PHCHR regularly when visiting the health centers, 79% reported that their own writing in the record was helpful, and 92% favored the PHCHR being permanently adopted. Use of the record did not influence the utilization of healthcare services, parents' knowledge of their child's health, or parents' satisfaction with information or communication with professionals.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Hagiwara, 2013 Palestine	Quasi-experimental	MCH treatment centers	Mothers who were expose and not exposed to the MCH Handbook	340	MCH handbook	Mothers who did not use the MCH Handbook	Knowledge related to MCH such as the importance of exclusive breastfeeding and how to cope with the risks of rupture of membranes during pregnancy increased among MCH handbook users, especially among less-educated women. The MCH handbook may be an effective tool for communication with health providers and husbands, for both highly educated and less-educated women during their first pregnancy.
Hamilton, 2012 Australia	Mixed methods	New South Wales (NSW)	Parents (mothers) who had at least one child aged between 0-4 years old	126	Child personal health record (CPHR)	No comparison group	CPHR can play an important role in communicating information regarding a child's health and development between parents and professionals, it is perhaps underutilised. Opportunities for use were reduced where there were dual systems in place, such as online records for immunization. Some information in the CPHR had the potential to escalate concerns about infant development. This was particularly the case for the growth charts, and it appeared that further explanation may have supported mothers and reduced their concerns. It was also the case that mothers did not pay attention to developmental indicators that they did not understand, such as head circumference.
Hampshire, 2004 UK	Cross-sectional	Nottingham	Mothers	401	Personal child health record (PCHR)	No comparison group	The PCHR is used by most mothers and is important for providing health promotion material to all families with young children. It may be particularly useful for first-time and teenage mothers.
Harrison, 1998 South Africa	Descriptive prospective study	17 child health clinics	Health personnel, mothers/caregivers	35 health personnel and 150 mothers/ caregivers	Road-to-Health (RTH) card	No comparison group	Most nurses supported the concept of an RTH card, but a large majority recommended that it be replaced with a notebook retained by the mother. A significant proportion of health personnel did not know how to use the weight-for-age chart. Most mothers attending clinics carried the card, but this number dropped for hospital visits and consultations with private doctors. Mother's understanding of the card was limited. For mothers, the weight-for-age chart, immunization schedule, and milestone section are obscure.
Higashiyama 2013 Japan	Qualitative case study	Hospital in the Kansai (Osaka) area	Couple adopting a baby	2	MCH handbook and nursing counseling	No comparison group	A case in which perinatal staff and medical social workers cooperated with a child guidance center to reduce the anxiety of adoptive parents and build good parent-child relationships for adoptive parents of special adoption. Nurses explained how to apply for a MCH handbook before the birth of their adopted child.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Hokama, 2000, Japan	Cross-sectional	Naha, Okinawa	Mothers of 3-5 month old children who have come for check-up	281	MCH handbook	No comparison group	Over 70% of mothers had read the pages on parenting. More than half of the mothers had filled in the pages of their child's development and growth chart. Reading and filling out the handbook were associated with maternal characteristics, with older mothers and mothers with little childcare experience filling out the handbook more. Over 90% of mothers replied that the information in the handbook was useful. The most highly evaluated pages were those on child health, growth and vaccination.
Ikeda, 2020 Japan	Cross-sectional	Japan	Foster parents	506	MCH handbook	No comparison group	The MCH handbook provided important information about the foster child. Though, inconvenience was noted for those without an MCH handbook and lack some birth information (e.g., birth weight, birthplace, blood type, etc.)
Jeffs, 1994 Australia	Quasi-experimental	New South Wales (NSW), Australia	Households with children aged four years or less and health care providers	1,533	Introduction of personal health records (PHR) since 1988	Five years after the introduction of personal health records	PHR was well retained, with 89% claimed retention at 4 years, and over 78% of parents able to produce the record for inspection at interview. Of the records examined, 91% had at least one immunization recorded while 68% had a complete regimen documented by age 4 years. Overall, 93% of parents expressed satisfaction with the PHR, while 64% of all health care providers also felt that the PHR was 'beneficial to the health care children received', although only 53% of them used it regularly to record their findings. It is concluded that the PHR currently issued in NSW is well retained and valued by parents, and used by and useful to a range of health professionals.
Lee, 2016 USA	Qualitative	Hospital	Spanish-speaking families and minority English-speaking families	40	Patient Passport	Usual care	The most common themes in the qualitative analysis of the interviews were: 1) organization of medical care; 2) emotional expressions about the hospitalization experience; and 3) overall understanding of the process of care. Spanish- and English-speaking families had similar patient satisfaction experiences, but the Passport families reported improved quality of communication with the medical care team.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Matsumoto, 1996 Japan	Quantitative case study	Teaching hospital in Nagoya	Post-partum women, first-time and second time mothers	210	MCH handbooks of 1st and 2nd generation mothers	No comparison group	Among the intervention group, 151 cases (71.9%) had seen or had received their MCH handbook when they were young, which was used by their mothers during pregnancy. However, the degree of utilization varied depending on the timing of when they had seen or received it. Utilization was high from those who had received the MCH handbook from their mothers. Regardless of the intervention or control, 174 cases (82.9) were considering giving their MCH handbook to their children and 76.4% (133 cases) were thinking that "marriage or pregnancy" was the best time. However, that awareness did not necessarily correlate with the self-filling status of the MCH handbook. To promote the intergenerational utilization of the MCH handbook, support for each period in the life cycle, including school health, is indispensable. The MCH handbook is a health guidance that can be passed on to future generations and used for a lifetime. By promoting the use of the MCH handbook book within the current generation, behaviors such as self-management of health, can be passed down to future generations.
McKinn, 2017 Vietnam	Qualitative	Tuan Giao District, Dien Bien Province	Thai and Hmong ethnic minority women who were currently pregnant or mothers of children under five in October 2015	37	MCH handbook	No comparison group	Ethnic minority women generally reported that health professionals delivered health information in a didactic, one-way style, and there was a reliance on written information (Maternal and Child Health handbook) in place of interpersonal communication. The health information they receive (both verbal and written) was often non-specific, and not context-adjusted for their personal circumstances. Women were therefore required to take a more active role in interpersonal interactions in order to meet their own specific information needs, but they are then faced with other challenges including language and gender differences with health professionals, time constraints, and a reluctance to ask questions.
McMaster, 1996 Bosnia and Herzegovina	Cross-sectional	Near Tuzla	Mothers and children in the collective centers and from the local community	571	Booklets (incorporating health records and health advice) were distributed to displaced and other families	No comparison group	Personal child health record and advice booklets not only provided essential data on immunization, nutrition, and prevalent medical disorders but also appeared to benefit the young population by supplying a permanent health record and health education material.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Minewaki, 2019 Japan	Qualitative case study	Public hospital in Kawasaki City, Kanagawa Prefecture	Mother who had previously experienced two early term miscarriages and was diagnosed with intrauterine fetal death (IUFD) at the 11th week of pregnancy	1	MCH handbook	No comparison group	Birth plan was realized according to the wishes of the mother and have the medical staff fill out the MCH handbook. The nurse who reflects on the experience tries to understand the grieving process of the mother by using Neimeyer's framework "those who experience the death of a loved one goes back and forth between the three phases of avoidance, assimilation, and adaptation. " and concludes that the mother was going back and forth from the assimilation phase. She thinks of how she could have better communicated but feels relieved when the mothers says, *"Thank you for holding the box as you would hold a sleeping baby when you brought the baby to me. Thank you for treating this child as a human being. By choosing the baby's clothes and hugging it, I was able to do something as it's mother. "*
Moore, 2000 UK	Quasi-experimental	Leicestershire county	Parents of British children who are likely to have special educational needs	99	Designed a record for disabled children as a supplement to the Leicestershire child health record. The intervention phase lasted 6 months. Only families in groups 1 and 3 received the new record.	Families who did not use the new record (Group 2)	Most of the entries were factual, and the principal use of the new document was as an aide-memoire. There was no evidence that the record improved the parent's perception of their child's general health care, nor that it contributed to the overall level of communication between parents and professionals.
Naito, 2019 Japan	Retrospective cohort	Community health center in Kurume City, Fukuoka	Pregnant women who submitted a pregnancy notification form in 2014.	2,986	MCH handbook	Those who were not registered and did not receive the MCH handbook	Being 35 years or older (OR[odds ratio]=1.41), height less than 158 cm (OR=1.45), non-pregnant body mass index (BMI) less than 18.5 (OR=1.48), and detection of physical abnormalities by a physician during the pregnancy (OR=2.20) were independent maternal factors that were significantly associated with low birth weight. Being aged 35 years or older (OR=2.05) and smoking (OR=3.42) were independent factors that were significantly associated with miscarriage and stillbirth. Also, the cessation of alcohol use (OR=0.51) significantly reduced this risk.
O'Flaherty, 1987 Australia	Prospective cohort	Maternity unit of Camden hospital	All mothers of babies who were born over one calendar month and health care providers	237	Personal health record	No comparison group	Eight per cent of mothers lost the records and three more said they had not been given a record while in hospital; a total of 10% of mothers had either lost or misplaced the record. There were no particular demographic characteristics which identified the mothers who were more likely to lose the record. Most parents liked personal health records and used them frequently, as did the community health staff. Most private doctors, however, did not find them useful. Before wider distribution of such records is contemplated health workers should be adequately prepared, especially doctors in the private sector.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Ogasawara, 2016 Japan	Cross-sectional	Great East Japan Earthquake disaster areas	Mothers, health and medical staff working in the disaster area	51	MCH handbook	No comparison group	The "vaccination record", "delivery situation", "1 month check-up" and other useful information were recorded. Iwate Prefecture's perinatal medical information system "Iiha-tobu" and the MCH handbook were useful during the disaster and utilized widely. For the MCH handbook to be able to survive future large disasters, efforts must be made to realize e-MCH handbook and for data to be kept in the cloud.
Ogawa, 2021 Japan	Cross-sectional	Four maternity facilities with labor and delivery rooms in Maebashi, Gunma	Pregnant women	1,009	MCH handbook	No comparison group	The MCH handbook provided disaster preparedness knowledge, especially among mothers who used the self-reporting sections of the MCH handbook.
Osaki, 2018 Indonesia	Cluster RCT	13 health centers in Garut district of rural Java, Indonesia	Pregnant women attending one of the selected health centers between 2007 and 2009	454	MCH handbook	Usual care	Respondents in the intervention area received consecutive MCH services including two doses of tetanus toxoid injections and antenatal care four times or more during pregnancy, professional assistance during child delivery and vitamin A supplements administration to their children, after adjustment for confounding variables and cluster effects (OR=2.03, 95% CI: 1.19–3.47). In the intervention area, home care (continued breastfeeding; introducing complementary feeding; proper feeding order; varied foods feeding; self-feeding training; and care for cough), perceived support by husbands, and lower underweight rates and stunting rates among children were observed.
Phipps, 2001 Australia	Qualitative	Home or antenatal appointment in hospital	Pregnant women	21	Woman-held maternity records	No comparison group	Maternal record holding had the potential to improve the level of communication between the health care worker and the pregnant woman and provided a greater sense of sharing and communication within the family. Woman's partner become better informed and more involved in the pregnancy.
Polnay, 1989 United Kingdom	Prospective cohort	Two largely working-class areas of Nottingham with large council estates	Mothers of all the children who were born from January to December 1983	67	Nottingham baby book	Non-user of Nottingham baby book	The book was well used by the majority of parents with 80% of parents having read all the booklet by the time their babies were three months old. Among the parents, 70% of them had retained the booklet when their children had reached the age of one year.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Seto, 2006 Japan	Qualitative case study	Public hospital in Iwamizawa City, Hokkaido Prefecture	Teenage mother and father	2	MCH handbook and kangaroo care	No comparison group	An 18-year-old woman underwent maternity checkup from the beginning of her pregnancy without any abnormalities. Around the 22nd week and 4 days of pregnancy, she was diagnosed with imminent preterm birth due to abdominal tension and vaginal bleeding, and was hospitalized. She delivered a boy, but doctors were not able to save his life. Even after active treatment was discontinued, there was a heartbeat and some breathing movement, therefore, the family spent time with the baby boy. After confirming the death, kangaroo care was continued for about an hour. After that, the baby was dressed in clothes that the family had prepared and a foot print and a handprint was taken as a token. When the mother discharged, the baby's name was written in the MCH handbook and words of gratitude for the birth of the child were written.
Shah, 1993 Multi-countries	Quasi-experimental	13 centers in eight countries (Egypt, India, Pakistan, Philippines, Senegal, Sri Lanka, Democratic Yemen, and Zambia)	The participating centers tested the HBMR in a variety of circumstances, such as literate and illiterate populations, different geographical and cultural conditions, and communities with easy or poor access to health services in rural and urban populations.	14,000 to 250,000	Home-based maternal record (HBMR)	Non-user of HBMR	The used of the HBMR had a favorable impact on utilization of health care services and continuity of the health care of women during their reproductive period. When adapted to local risk conditions, the HBMR succeeded in promoting self-care by mothers and their families. The introduction of the HBMR increased the diagnosis and referral of at-risk pregnant women and newborn infants, improved family planning and health education, led to an increase in tetanus toxoid immunization, and provided a means of collecting health information in the community. The HBMR was liked by mothers, community health workers and other health care personnel. Mothers became more involved in looking after their own health and that of their babies. The training and involvement of health personnel from the start of the HBMR scheme influenced its success in promoting maternal and child health care. It also improved the collection of community-based data and the linking of referral networks.
Shimizu, 2007 Dominican Republic	Cross-sectional	Dajabón	Mothers who received the MCH Handbook and children under the age of 5 using the handbook	6,633	MCH handbook	No comparison group	The evaluation and regular monitoring visits revealed positive results: as for pregnant women, the handbooks were well accepted for their friendliness, simplicity, durability and mobility, and the rate of their receiving antenatal and postpartum cares at designated clinics or hospitals increased; as for newborns and children, the immunization coverage improved while common problems such as diarrhea decreased; and as for health personnel, the handbook helped clarify the division of work and enhanced their sense of responsibility, communication, and continuity and integration of service.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Sugi, 1985 Japan	Cross-sectional	Health check-up stations	Caregivers of 18-month-old children	111	MCH handbook	No comparison group	Interest in the MCH handbook was higher at check-up time compared to consultation time, for both the medical health care workers and the caregivers. The section which was of least interest was child and maternal oral hygiene. The page which was read the most was nutrition during pregnancy. However, the page on financial support and subsidies for maternal and child medical care was the least read. About 63.2% of mothers made notes (recorded) with those who were pregnant with their first child and was not working more likely to record the process. Item-wise, names of the parents, birth certificate record, due date and other items to be filled out by the pregnant woman, as well as the first month. Extra notes, dental records up until 18 months, timing of restart of menstruation and other post-natal maternal records were less likely to have been filled out.
Takeda, 2002 Japan	Cross-sectional	A city in Okinawa Prefecture	Caregivers of 18-month-old children	230	MCH handbook	No comparison group	Most mothers read the vaccination page (85.8%), information on childcare (77.1%), and accident prevention (76.2%). However, only 33.4% out of those who replied that they read the handbook and read the Children's Charter. About 90% of those who replied that the handbook was useful, replied that the information on the vaccination page helped eliminate worries, 88.8% said the information on childcare was useful, 87.1% said that the information helped eliminate worries on her child's health and growth. No significant association was identified between those who read the handbook, those who accepted the utility of vaccination and the mother's age, schooling, maternal employment and child rank.
Tanabe, 2011 Japan	Multi-facility cohort study	Four out of five delivery facilities within Sendai City, Miyagi Prefecture	First generation and current generation mothers	724	MCH handbooks of current generation mothers	MCH handbooks of first-generation mothers	Using the MCH handbook, the associations of anthropometric factors and course of pregnancy and delivery comparisons between the two generations were evaluated. The study found some associations between a mother's course of pregnancy and delivery and her daughter's. The data showed a significant and positive association in: height, weight, and body mass index (BMI) before pregnancy, weight gain during pregnancy, systolic and diastolic blood pressure in both second and third trimester, baby's weight and head circumference. Birth weight of offspring was more associated with mother's birth weight than BMI before pregnancy and weight gain during pregnancy. This suggests that the research of a mother's course of pregnancy and delivery could offer some predictions concerning her daughter's pregnancy and delivery.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Umeda, 2015 Mongolia	Cross-sectional	Zavkhan Prefecture, Mongolia (1100 km west of Ulaanbaatar)	Mothers and medical workers of Zavkhan General Hospital and village health center	42	MCH handbook	No comparison group	Of 42 health providers, 66% used the mother and child handbook as a medical record, 57% used it as a communication tool with mothers, 50% saw the mother handbook as an individual record to record the fetus growth, and a textbook or guidebook on childcare support, 45% saw the handbook as a tool to promote participation in childcare for fathers and 28% saw the handbook as a tool to nurture the next future generation's parents. One respondent wrote that there should be a space for the doctor to write advice instead of just providing information. Another wrote that the handbook should have a space where advice for the father could be written in. What could be done to support his wife and should include information on tobacco and alcohol so that the husband and family could be more attentive to the health of mother-child.
Walton, 2007 UK	Cross-sectional	10 child health clinics located in two primary care trusts; one in central London and the other in Buckinghamshire in July 2004	Parents who arrived at the clinic with new PCHR	89	New Personal Child Health Record (PCHR)	No comparison group	Nearly all parents (98%) reported that they used the PCHR as a record of their child's health and development and 92% reported that they 'always' took it with them when seeing healthcare staff about their child. Some parents (22%) indicated that they had not been given a satisfactory explanation as to how to use the PCHR, at the time it was issued to them. Parents reported that health visitors were more likely than other health professionals to use the PCHR both to obtain information about their child and to record information. The majority of respondents (78%) were happy for the level of maternal education to be documented in their child's PCHR.
Whitford, 2014 Scotland	Qualitative	Two National Health Service Board regions in northeast Scotland	Pregnant women (after 34 weeks) and if they agreed, at about eight weeks postnatally.	42	Birth plan within woman-held maternity records	No comparison group	Staff and women were generally positive about the provision of the birth plan section within the record. Perceived benefits included the opportunity to highlight preferences, enhance communication, stimulate discussions, and address anxieties. However, not all women experienced these benefits or understood the birth plan's purpose. Some were unaware of the opportunity to complete it or could not access the support they needed from staff to discuss or be confident about their options. Some were reluctant to plan too much. Staff recognized the need to support women with birth plan completion but noted practical challenges to this.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Wright, 2005 UK	Prospective cohort	One district in England (Gateshead)	Mothers of all babies born between June 1, 1999 to August 31, 2000	1,369	Personal Child Health Record (new and old)	No comparison group	Parents rated both record types highly and the majority used them regularly to take to baby clinics and for information. Health visitors wrote frequently in the record, compared with only half of parents and less than a quarter of family doctors. Old format records were significantly more likely to be taken to and written in by the family doctor. Parents used new format records less as a source of information, but were no more likely to use other recommended information sources. Parents with new format records showed better recall of information found only, or more prominently in the new records, but the actual differences were small.
Yahata, 2005 Japan	Qualitative	Akita prefecture	Parents of non-measles vaccinated children	9	MCH handbook	No comparison group	Caregivers were not against measles vaccination (positive attitude) The main reasons why they had not vaccinated their child against measles were "My child caught a cold, and it was difficult to find time afterwards", "I also intend to go vaccinate my child but can not seem to get there", "I don't have time to go for vaccination". In order to raise vaccination coverage rate, caregivers proposed clearer messaging on "measles vaccination safety in the MCH handbook" and information that "Vaccination can be done even outside your local burrough", or other information such as "If measles vaccination dates were fixed, I would do everything to get my child vaccinated then".Others also said that the health administrators should play a more active role such as "Getting health workers to flag that measles vaccination has not been done at child health days".
Young, 1990 USA	Qualitative	Federally funded clinic open year round	Infants and preschool-age children who received well-child services at Tri-County Community Health Center	560	Growth chart	No comparison group	Professional staff consistently reported that the record was a useful aid in teaching migrant parents about their children's growth. Parents receiving the records appeared more attentive and receptive to nutrition counseling. They also asked more questions and volunteered more pertinent information about their children. Including a photo of the child also distinguishes these from other versions of family-carried records.

Supplementary file 4 (continued)

Study	Study design	Study setting	Study population	Sample size	Intervention	Comparator	Reported outcomes
Yuge, 2010 Japan	Cross-sectional	Health check-up stations	Mothers of four-month-old, 18-month-old and three-year old children who have come for check-up	321	MCH handbook	No comparison group	Utility point average was 3.4-3.5. There was no difference between child age and mother and child health status. Mothers found the pages which medical workers filled out useful. These were "delivery record", "vaccination record" and "neonatal record" pages. There were very few childcare instruction items/pages which were useful. Mothers with previous children found the pages "experience of seeing the MCH handbook during childhood", "discuss the handbook", "received explanations from the pediatrician using the handbook" more useful than first-time mothers. Average points on the whether mothers wanted to show the handbook to their children, on continuity was 4.5-4.8 points, mothers with 4 month old children had a higher continuity awareness than 3 year old children. Mothers who had seen their own handbook when younger had a higher continuity awareness than those who had not. There is a statistically significant association between those who see utility in the handbook and handing over the handbook to their children.